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# Beyond Linear Concepts: Discovering and Aligning Non-Linear Concept Manifolds in Large Language Models

*Modelling Concepts as Low-Dimensional Riemannian Manifolds Reveals Consistent Block Structures and a Syntax-to-Semantics Transition Universal Across Architectures*

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Adapting Non-Linear Multi-Dimensional Concept Discovery (NLMCD) from computer vision to token-level LLM activations and introducing the Concept-Based Alignment (CBA) score, this work reveals that concept organization in LLMs exhibits consistent block structures across layers and architectures, with representations staying syntax-dominated through 70% of the network before transitioning to mixed syntactic-semantic structure in late layers—patterns invisible to linear methods.

## AT A GLANCE

**Problem:** Existing MI methods assume linear concept representations, missing the non-linear geometric structure of LLM activations.

**Why it matters:** Linear methods may produce incomplete or misleading interpretations of how concepts are organized in large models.

**Method:** NLMCD fits low-dimensional manifolds to token activations; CBA score (generalized Rand index) compares manifolds across layers and models without requiring explicit feature matching.

**Result:** Consistent block structures at intermediate layers; syntax-dominated through 70% of network then mixed syntactic-semantic; CBA is 15–30% more sensitive than PCA or CKA.

**Evidence:** Five LLM architectures (Llama, Mistral, Falcon, and others) on Wikipedia + code corpora.

## The Big Picture

Mechanistic interpretability seeks to decode the internal representations of large language models—how concepts are organized, transformed, and composed across layers. Dominant methods (probes, sparse autoencoders, dictionary learning) rest on a shared linearity assumption: each

concept occupies a direction in activation space, and its presence is measured by a linear classifier or inner product.

This assumption is increasingly challenged by empirical evidence showing curved, non-linearly distributed activation clusters. If concepts genuinely occupy low-dimensional manifolds rather than directions, linear methods will systematically miss or mischaracterize them. This paper takes the manifold hypothesis seriously and develops tools to act on it, revealing organizational principles in LLMs that linear baselines cannot detect.

## Why Existing Approaches Fall Short

**Linear probes** measure concept presence as a dot product with a learned direction, entirely missing curved or non-linearly distributed concept regions.

**Sparse autoencoders (SAEs)** decompose activations into sparse linear combinations; each dictionary feature is a direction, and compositions are linear sums. Non-linear manifold structure is encoded only incidentally.

**CKA and PCA-based comparisons** compare linear subspaces across layers using canonical angles; they are insensitive to non-linear geometry and cannot detect manifold-level structural differences.

**No cross-model alignment:** Existing methods lack a rigorous way to compare concept organization across different architectures without explicit feature matching, which assumes a known bijection between concepts—not generally available.

## Core Mechanism

**NLMCD on LLM Activations.** For each concept category  $c$  (syntactic POS tag, dependency role, semantic entity type) and each layer  $l$ , the paper collects token activation sets  $\mathcal{H}_c^l$  and fits a low-dimensional Riemannian manifold  $\mathcal{M}_c^l$  using a neural network atlas.

**Concept-Based Alignment (CBA) Score.** CBA measures whether two manifolds partition the same token set into the same neighborhoods, without requiring explicit feature matching:

$$\text{CBA}(\mathcal{M}_c^l, \mathcal{M}_c^{l'}) = \text{RandIndex}(G_c^l, G_c^{l'})$$

where  $G_c^l$  is a neighborhood graph on tokens  $\mathcal{T}_c$  derived from geodesic distances on  $\mathcal{M}_c^l$ . CBA is 1.0 when both manifolds partition the token space identically and 0.5 at chance.

## Technical Formulation

Let  $\mathbf{h}_l^{(t)} \in \mathbb{R}^d$  be the activation of token  $t$  at layer  $l$ . The NLMCD atlas  $\phi_c^l : \mathbb{R}^k \rightarrow \mathbb{R}^d$  (with  $k \ll d$ ) is fitted by:

$$\min_{\phi} \sum_{t \in \mathcal{T}_c} \|\mathbf{h}_l^{(t)} - \phi(z^{(t)})\|^2 + \lambda \text{Reg}(\phi)$$

where  $z^{(t)} \in \mathbb{R}^k$  are latent coordinates learned jointly, and  $\text{Reg}(\phi)$  is a smoothness regularizer.

The CBA between concept  $c$  at layers  $l$  and  $l'$  uses the generalized Rand index:

$$\text{CBA} = \frac{|\{(i, j) : \text{near in } G_c^l \Leftrightarrow \text{near in } G_c^{l'}\}|}{|\mathcal{T}_c|^2}$$

All-pairs CBA matrices across layers  $\{1, \dots, L\}$  reveal which layer intervals share similar concept geometry.

### Learning or Inference Procedure

#### Manifold fitting (offline):

1. Run the LLM on a large corpus ( $\approx 10\text{M}$  tokens).
2. Extract activations  $\{\mathbf{h}_l^{(t)}\}$  for all tokens and layers.
3. For each concept category  $c$  and layer  $l$ , fit NLMCD atlas  $\phi_c^l$  using stochastic gradient descent.

#### CBA computation (post-fitting):

1. Compute pairwise geodesic distances on each fitted manifold.
2. Build neighborhood graphs  $G_c^l$  and compute Rand index between all (layer, model) pairs.
3. Visualize as CBA alignment matrices to identify block structures.

### What the Guarantee Says

NLMCD provides a consistent manifold estimate as sample size grows, under standard regularity conditions on manifold geometry. CBA is a metric on the space of neighborhood partitions. The main empirical claim—cross-architecture consistency of block structures—is validated by replicating CBA patterns across five model families including architectures trained on different data regimes.

### Experimental Findings

- **Block structures:** CBA matrices show consistent high-alignment blocks at intermediate layers ( $\approx$  layers 10–20 in a 32-layer model).
- **Syntax-to-semantics:** Syntactic concepts dominate CBA structure through  $\approx 70\%$  of network depth; semantic concepts become prominent in the final 30% of layers across all tested models.
- **Cross-architecture:** Block structures are consistent across Llama, Mistral, Falcon and two additional

families at equivalent depth fractions.

- **CBA sensitivity:** CBA is 15–30% more sensitive than PCA-based or CKA-based baselines, as measured by detection of synthetic perturbations.
- **Non-linearity:** Linear probes in place of NLMCD miss the block transitions entirely, confirming that non-linear geometry carries genuine structural information.

### Ablations and Interpretation

**Manifold dimension  $k$ :** Values  $k \in \{2, 4, 8\}$  all produce similar CBA patterns;  $k = 4$  offers the best fidelity-cost trade-off.

**Corpus composition:** Code-heavy vs. text-heavy corpora show similar syntactic block patterns but differ in semantic concept structure of late layers, aligned with known training data differences.

**Tokenization:** Character-level tokens show less pronounced block structure than subword tokens, suggesting the patterns are tightly coupled to the tokenization scheme.

### Reference

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